

ASHEBORO, NC, USA

WMO: 746935

Lat: 35.654N

Lon: 79.895W

Elev: 673

StdP: 14.34

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 03709

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.1	23.5	-0.4	5.5	29.2	3.8	6.9	30.7	18.5	56.5	16.7	53.8	2.8	350	0.363

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.1	93.0	73.5	90.9	73.9	89.5	73.7	78.2	85.6	76.9	85.5	75.8	84.2	4.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.5	136.9	81.3	74.5	132.4	80.3	73.2	126.4	79.1	42.2	86.4	40.6	85.8	39.7	84.2	
(p)															

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.1	12.9	11.4	13.9	96.4	5.0	3.2	10.3	98.7	7.4	100.6	4.6	102.4	1.0	104.7		
(5)			11.6	80.4	4.7	2.7	8.2	82.4	5.4	84.0	2.7	85.5	-0.7	87.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	60.7	41.2	44.2	51.3	60.7	68.2	75.6	78.8	77.6	71.8	61.3	51.2	45.2		
	DBStd	15.31	10.56	9.35	9.72	8.27	6.83	5.13	3.94	4.50	5.62	8.11	8.77	9.18		
	HDD50	946	312	211	106	12	0	0	0	0	0	8	89	208		
	HDD65	3206	739	585	434	177	50	2	0	0	13	169	421	616		
	CDD50	4846	39	48	146	334	566	769	892	855	653	360	125	59		
	CDD65	1629	1	2	9	48	151	321	427	390	216	55	7	2		
	CDH74	13818	3	7	73	447	1177	2835	3914	3364	1569	376	47	6		
	CDH80	5051	0	0	6	88	307	1103	1630	1345	492	77	3	0		
(14) Wind	WSAvg	3.8	4.5	4.7	5.0	5.2	4.0	3.2	2.9	2.5	3.0	3.2	3.6	3.8		
(15) Precipitation	PrecAvg	45.3	3.5	3.1	4.2	3.6	3.4	4.1	4.7	4.3	4.4	3.7	3.4	3.2		
	PrecMax	61.6	5.7	5.9	8.3	7.5	8.5	9.2	9.5	8.0	10.5	13.3	8.2	5.8		
	PrecMin	32.6	1.1	1.3	1.6	0.7	1.0	0.8	2.1	1.7	0.6	0.0	0.6	1.5		
	PrecStd	6.7	1.5	1.2	1.7	1.8	1.8	2.2	2.0	1.5	2.8	3.1	2.0	1.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	71.6	74.8	80.9	87.7	90.0	94.9	96.8	97.2	91.3	86.4	79.3	73.3		
		MCWB	62.5	61.9	61.3	63.8	70.9	72.8	73.8	73.8	70.7	71.9	64.7	63.4		
	2%	DB	66.1	69.8	75.3	82.2	86.3	91.2	92.9	92.8	88.4	81.6	73.3	68.8		
		MCWB	58.6	58.1	60.4	64.5	69.1	73.8	74.4	73.8	70.5	67.9	62.3	62.8		
	5%	DB	63.0	64.4	72.3	79.5	83.4	89.7	90.8	90.3	85.7	78.7	70.5	64.4		
		MCWB	55.9	55.3	58.0	62.7	68.6	73.2	75.0	74.6	70.7	66.6	60.8	59.0		
	10%	DB	58.0	60.7	67.9	75.4	81.1	86.6	88.6	87.7	82.2	74.6	66.1	61.2		
		MCWB	50.8	51.7	56.8	61.8	67.2	72.3	74.4	73.7	70.0	66.2	58.6	55.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.4	64.7	66.8	72.3	75.2	82.2	83.1	82.1	77.3	74.5	70.5	67.2		
		MCDB	68.7	70.8	71.4	78.6	82.6	85.1	88.4	86.1	83.7	79.2	76.1	69.6		
	2%	WB	60.9	61.8	63.4	68.5	72.2	77.2	78.0	77.6	75.3	72.8	66.2	64.1		
		MCDB	64.9	66.6	70.6	76.4	79.8	84.6	86.9	86.4	81.9	77.5	70.4	67.2		
	5%	WB	57.4	58.0	61.3	66.5	70.8	75.3	76.8	76.6	73.9	70.3	63.3	60.7		
		MCDB	61.9	63.7	68.4	74.9	78.6	84.4	86.1	85.1	80.2	75.7	68.0	64.0		
	10%	WB	52.5	53.2	59.2	64.2	69.4	74.0	75.7	75.5	72.4	67.4	59.4	56.2		
		MCDB	56.8	58.2	66.2	71.8	77.3	82.8	84.6	83.5	78.5	73.2	65.4	60.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	18.4	19.3	20.9	21.7	19.9	18.7	18.1	17.6	18.2	20.1	21.6	18.5		
		MCDBR	22.0	24.0	23.8	25.2	22.4	21.3	20.5	20.9	22.7	22.2	24.4	20.9		
		MCWBR	16.2	15.4	12.6	12.0	9.6	8.2	7.9	7.3	8.0	11.2	15.3	14.8		
	5% WB	MCDBR	19.3	21.1	21.3	21.9	18.7	18.9	18.5	17.5	16.7	18.0	19.6	17.4		
		MCWBR	16.6	16.8	15.0	13.6	9.3	9.5	8.4	8.1	9.2	10.9	15.7	14.3		
(40) Clear-Sky Solar Irradiance	taub		0.311	0.324	0.359	0.398	0.449	0.484	0.533	0.513	0.422	0.383	0.328	0.310		
	taud		2.504	2.460	2.397	2.288	2.182	2.120	2.006	2.047	2.296	2.376	2.495	2.536		
	Ebn at Noon		283	292	289	281	267	257	243	245	265	266	274	276		
	Edh at Noon		26	30	35	42	47	50	56	52	39	33	26	24		
(44) All-Sky Solar Radiation	RadAvg		820	1029	1366	1711	1897	2027	1898	1730	1453	1187	903	705		
	RadStd		53	126	96	129	154	148	123	83	130	155	72	61		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (66 neighbors)	+0.76	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+265	+187

Nomenclature: See separate page